Unit 4: Further Mechanics, Fields and Particles

IA2 compulsory unit

Externally assessed

4.1 Unit description

Introduction	This topic covers further mechanics, electric and magnetic fields, and nuclear and particle physics. This topic may be studied using applications that relate to mechanics, for example transportation and fields, for example communications and display techniques. This topic also enables students to develop practical and mathematical skills.
Practical skills	In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating the effect of mass, velocity and radius of orbit on centripetal force, using a coulomb meter to measure charge stored and using an electronic balance to measure the force between two charges.
Mathematical skills	Mathematical skills that could be developed in this topic include translating between degrees and radians and using trigonometric functions, sketching relationships that are modelled by $y = k/x$, and $y = k/x^2$, using logarithmic plots to test exponential and power law variations, interpreting logarithmic plots and sketching relationships that are modelled by $y = e^{-x}$.